

Biology Unit 3: Cell Membranes and Transport

The plasma membrane

Every cell is enclosed by a plasma membrane that controls what enters and what leaves. The plasma membrane is a phospholipid bilayer: the hydrophilic heads face the water on both sides and the hydrophobic tails point inward. That arrangement is not decorative — it is exactly what makes the membrane a barrier in the first place.

Because the interior of the bilayer is oily rather than watery, it does not treat every molecule the same way. The nonpolar interior of the bilayer blocks most polar molecules and all ions from crossing on their own. Size alone does not decide the outcome; polarity does.

Some molecules do cross without any assistance at all. Small nonpolar molecules such as oxygen and carbon dioxide diffuse straight through the bilayer. Others cannot manage that on their own. Ions and large polar molecules such as glucose cross only through transport proteins. Those proteins are what the rest of this unit is about.

Passive transport: diffusion and osmosis

The simplest way anything crosses a membrane is diffusion. Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration. Because it is driven by the random motion of the particles themselves, diffusion requires no energy from the cell. The particles are not moving on purpose, but the statistics of many random collisions produce a net flow.

Water follows the same idea, with its own name. Osmosis is the diffusion of water across a selectively permeable membrane. Like ordinary diffusion, it runs downhill: water moves toward the side with the higher solute concentration, where the water concentration is lower. The solute itself may be unable to cross the membrane at all; it is still the water that moves in response.

This has visible consequences for a cell. In a hypotonic solution the cell gains water and swells; an animal cell may burst. In a hypertonic solution the cell loses water and shrinks. Between those two extremes, an isotonic solution keeps the cell's volume roughly stable, because water enters and leaves at equal rates.

Facilitated diffusion and active transport

Not every molecule that crosses the membrane down its gradient can do so unassisted. Facilitated diffusion uses channel or carrier proteins, but it is still passive: the molecules move down their gradient and the cell spends no energy. A protein is doing the work of providing a path, not the work of pumping. Water itself often crosses this way: aquaporins are channel proteins that speed the movement of water.

Moving a substance the other way, against its gradient, is a different matter entirely. Active transport moves substances against their concentration gradient, from low concentration to high, and requires energy from ATP. That energy dependence is the tell: if a cell runs out of ATP, its pumps stop and the gradients they maintain slowly dissipate.

The clearest example is the sodium-potassium pump. Each cycle of the sodium-potassium pump moves three sodium ions out of the cell and two potassium ions in, at the cost of one ATP. That leaves the inside of the cell slightly more negative than the outside. Nerve and muscle cells depend on the gradients this pump maintains.

Bulk transport and review

Some cargo is too large for any channel or carrier protein to handle one molecule at a time, so the cell moves it in bulk instead, wrapped in a piece of membrane. In endocytosis the membrane folds inward around the cargo and pinches off to form a vesicle inside the cell. This is how cells take in large particles, fluid, or even other cells.

The reverse process works the same way in the opposite direction. In exocytosis a vesicle fuses with the plasma membrane and releases its contents outside the cell. This is how cells secrete hormones, neurotransmitters, and waste. Both directions reshape the membrane itself and cost the cell energy, which is why bulk transport, like active transport, is never considered passive.

Across the whole unit, the throughline is the same: whether a substance crosses freely, needs a channel or carrier, or needs a pump spending ATP depends on its size, its polarity, and which direction it is trying to go relative to its own concentration gradient.